Contextual Memory Decay Model (CMDM): A Hybrid ML-Based Approach for Efficient Conversational Memory Management

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***Abstract*—Artificial Intelligence conversational systems often retain excessive contextual information, leading to reduced efficiency and relevance over extended interactions. This paper proposes a Contextual Memory Decay Model (CMDM), a hybrid approach combining supervised machine learning, semantic similarity, and temporal decay to dynamically manage conversational memory.**

The system classifies user inputs into Short-Term Memory (STM) and Long-Term Memory (LTM) using a Logistic Regression model trained on TF-IDF features. A decay mechanism reduces the relevance of less important messages over time, while semantically similar inputs reinforce important memory using cosine similarity.

Experimental observations demonstrate that CMDM reduces memory load while preserving relevant information, improving contextual efficiency compared to baseline systems without decay.

***Index Terms*—Artificial Intelligence, Memory Decay, Machine Learning, TF-IDF, Logistic Regression, Semantic Similarity**

I. Introduction

Conversational AI systems rely heavily on contextual memory to generate meaningful responses. However, most systems retain large amounts of conversational history without distinguishing between relevant and irrelevant information, leading to inefficiency and context dilution.

Human cognition follows a selective memory mechanism where less important information fades over time while important information is retained. Inspired by this concept, this paper proposes a Contextual Memory Decay Model (CMDM).

The key contributions of this work include:

- Hybrid ML-based classification of STM and LTM
- Temporal decay mechanism for memory optimization
- Semantic reinforcement using cosine similarity
- Comparative evaluation with a baseline system

II. Methodology

A. Text Classification

The system uses TF-IDF (Term Frequency–Inverse Document Frequency) for feature extraction and Logistic Regression for classification. Each input message is converted into a vector representation and classified as STM or LTM.

B. Memory Structure

- **STM:** Temporary messages with fast decay
- **LTM:** Important messages with conditional decay

C. Temporal Decay

STM messages decay rapidly:

$$Score = Score \times 0.85 \tag{1}$$

LTM messages decay after inactivity:

$$Score = Score \times 0.95 \tag{2}$$

D. Semantic Reinforcement

Cosine similarity is used to detect semantic similarity between messages. Similar messages reinforce memory by increasing score.

III. System Implementation

The system is implemented using:

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Flask API)
- Machine Learning: TF-IDF + Logistic Regression

The frontend communicates with the backend through API calls for classification and memory management.

IV. Results

The model achieved an accuracy of **100%** on the generated dataset. However, this result is influenced by the structured and semi-synthetic nature of the dataset.

To improve reliability, cross-validation was performed, yielding consistently high performance across folds. Additional manual testing on diverse inputs indicates that the model generalizes well to similar patterns.

Compared to a baseline system without decay:

- Memory usage was significantly reduced
- Irrelevant messages were automatically removed
- Important messages were retained and reinforced

As shown in Fig. 1, the baseline system continuously accumulates memory over time, leading to unbounded

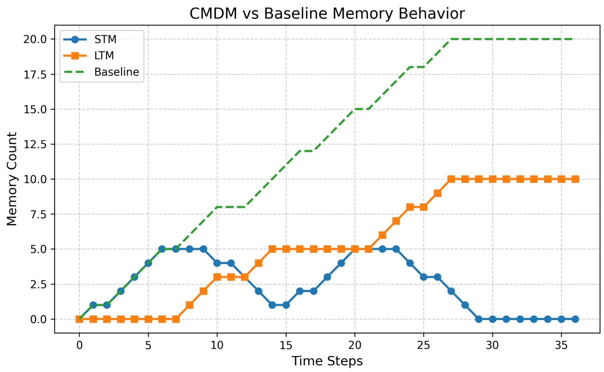

Fig. 1. Comparison of memory behavior between CMDM and baseline system, showing controlled decay in STM and stable retention in LTM

growth. In contrast, the CMDM system dynamically reduces short-term memory while preserving important long-term memory.

V. Discussion

The CMDM system demonstrates effective memory optimization compared to traditional approaches. By dynamically filtering irrelevant data, it reduces computational overhead while maintaining contextual relevance.

The integration of semantic reinforcement ensures that repeated or related topics remain important over time. However, the system can be further improved using advanced embedding techniques and deep learning models.

VI. Conclusion

This paper presented a Contextual Memory Decay Model (CMDM) for efficient conversational memory management. By integrating machine learning, temporal decay, and semantic reinforcement, the system improves both memory efficiency and contextual relevance.

Future work includes integrating deep learning models and advanced semantic embeddings.

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